

Annex 179C

(normative)

MDIs for 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8

179C.1 Overview

This annex defines the Medium Dependent Interface (MDI) for 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8. The MDI couples the PMD (specified in 179.8 and 179.9) to the cable assembly (specified in 179.11). The PMDs supportable for each MDI connector are given in Table 179C–1.

In all MDIs, the plug connector is used on the cable assembly and the receptacle is used on the PMD.

Table 179C–1—Number of PMDs supportable for each connector type

MDI connector type	200GBASE-CR1	400GBASE-CR2	800GBASE-CR4	1.6TBASE-CR8	Reference
QSFP224	1, 2, 4	1, 2	1	—	179C.2.1
QSFP-DD1600	1, 2, 4, 8	1, 2, 4	1, 2	1	179C.2.2
OSFP1600	1, 2, 4, 8	1, 2, 4	1, 2	1	179C.2.3

For 200GBASE-CR1, the mechanical interface between the PMD and the cable assembly is a mated pair of connectors meeting the requirements of 179C.2.1 (QSFP224), 179C.2.2 (QSFP-DD1600), or 179C.2.3 (OSFP1600).

For 400GBASE-CR2, the mechanical interface between the PMD and the cable assembly is a mated pair of connectors meeting the requirements of 179C.2.1 (QSFP224), 179C.2.2 (QSFP-DD1600), or 179C.2.3 (OSFP1600).

For 800GBASE-CR4, the mechanical interface between the PMD and the cable assembly is a mated pair of connectors meeting the requirements of 179C.2.1 (QSFP224), 179C.2.2 (QSFP-DD1600), or 179C.2.3 (OSFP1600).

For 1.6TBASE-CR8, the mechanical interface between the PMD and the cable assembly is a mated pair of connectors meeting the requirements of 179C.2.2 (QSFP-DD1600) or 179C.2.3 (OSFP1600).

The MDI connector electrical performance is consistent with the signal quality and electrical requirements of 179.9 and 179.11.

An MDI connector type may be used by one or more PMDs. The assignment of PMD signals to connector signals is specified in Table 179C–2, where as an example 0:DL0n refers to the DL0n signal of the first PMD; see 179.8.2 and 179.8.3 for signal naming definitions. When not all connector contacts in an MDI connector are physically connected to PMD signals, the connector contacts corresponding to lower PMD numbers in Table 179C–2 shall be used.

Table 179C–2—PMD to connector signal assignments

PMD signal <PMD number>:<PMD signal>				Connector signal
200GBASE-CR1	400GBASE-CR2	800GBASE-CR4	1.6TBASE-CR8	
0:DL0n	0:DL0n	0:DL0n	0:DL0n	DL0n
0:DL0p	0:DL0p	0:DL0p	0:DL0p	DL0p
1:DL0n	0:DL1n	0:DL1n	0:DL1n	DL1n
1:DL0p	0:DL1p	0:DL1p	0:DL1p	DL1p
2:DL0n	1:DL0n	0:DL2n	0:DL2n	DL2n
2:DL0p	1:DL0p	0:DL2p	0:DL2p	DL2p
3:DL0n	1:DL1n	0:DL3n	0:DL3n	DL3n
3:DL0p	1:DL1p	0:DL3p	0:DL3p	DL3p
4:DL0n	2:DL0n	1:DL0n	0:DL4n	DL4n
4:DL0p	2:DL0p	1:DL0p	0:DL4p	DL4p
5:DL0n	2:DL1n	1:DL1n	0:DL5n	DL5n
5:DL0p	2:DL1p	1:DL1p	0:DL5p	DL5p
6:DL0n	3:DL0n	1:DL2n	0:DL6n	DL6n
6:DL0p	3:DL0p	1:DL2p	0:DL6p	DL6p
7:DL0n	3:DL1n	1:DL3n	0:DL7n	DL7n
7:DL0p	3:DL1p	1:DL3p	0:DL7p	DL7p
0:SL0n	0:SL0n	0:SL0n	0:SL0n	SL0n
0:SL0p	0:SL0p	0:SL0p	0:SL0p	SL0p
1:SL0n	0:SL1n	0:SL1n	0:SL1n	SL1n
1:SL0p	0:SL1p	0:SL1p	0:SL1p	SL1p
2:SL0n	1:SL0n	0:SL2n	0:SL2n	SL2n
2:SL0p	1:SL0p	0:SL2p	0:SL2p	SL2p
3:SL0n	1:SL1n	0:SL3n	0:SL3n	SL3n
3:SL0p	1:SL1p	0:SL3p	0:SL3p	SL3p
4:SL0n	2:SL0n	1:SL0n	0:SL4n	SL4n
4:SL0p	2:SL0p	1:SL0p	0:SL4p	SL4p
5:SL0n	2:SL1n	1:SL1n	0:SL5n	SL5n
5:SL0p	2:SL1p	1:SL1p	0:SL5p	SL5p
6:SL0n	3:SL0n	1:SL2n	0:SL6n	SL6n
6:SL0p	3:SL0p	1:SL2p	0:SL6p	SL6p
7:SL0n	3:SL1n	1:SL3n	0:SL7n	SL7n
7:SL0p	3:SL1p	1:SL3p	0:SL7p	SL7p

The MDI connectors shall have data signal and signal ground contact assignments as specified in Table 179C–3, and Table 179C–4 for the paired transmit and receive lanes (e.g., (SL0, DL0), (SL1, DL1)).

Table 179C–3—MDI connector contact mapping for QSFP224 and QSFP-DD1600

QSFP224	QSFP-DD1600	Connector signal name	Description
1	1	GND	Ground
2	2	SL1n	Transmitter inverted data input
3	3	SL1p	Transmitter non-inverted data input
4	4	GND	Ground
5	5	SL3n	Transmitter inverted data input
6	6	SL3p	Transmitter non-inverted data input
7	7	GND	Ground
13	13	GND	Ground
14	14	DL2p	Receiver non-inverted data output
15	15	DL2n	Receiver inverted data output
16	16	GND	Ground
17	17	DL0p	Receiver non-inverted data output
18	18	DL0n	Receiver inverted data output
19	19	GND	Ground
20	20	GND	Ground
21	21	DL1n	Receiver inverted data output
22	22	DL1p	Receiver non-inverted data output
23	23	GND	Ground
24	24	DL3n	Receiver inverted data output
25	25	DL3p	Receiver non-inverted data output
26	26	GND	Ground
32	32	GND	Ground
33	33	SL2p	Transmitter non-inverted data input
34	34	SL2n	Transmitter inverted data input
35	35	GND	Ground
36	36	SL0p	Transmitter non-inverted data input
37	37	SL0n	Transmitter inverted data input

**Table 179C–3—MDI connector contact mapping for QSFP224 and QSFP-DD1600
(continued)**

QSFP224	QSFP-DD1600	Connector signal name	Description
38	38	GND	Ground
—	39	GND	Ground
—	40	SL5n	Transmitter inverted data input
—	41	SL5p	Transmitter non-inverted data input
—	42	GND	Ground
—	43	SL7n	Transmitter inverted data input
—	44	SL7p	Transmitter non-inverted data input
—	45	GND	Ground
—	51	GND	Ground
—	52	DL6p	Receiver non-inverted data output
—	53	DL6n	Receiver inverted data output
—	54	GND	Ground
—	55	DL4p	Receiver non-inverted data output
—	56	DL4n	Receiver inverted data output
—	58	GND	Ground
—	59	DL5n	Receiver inverted data output
—	60	DL5p	Receiver non-inverted data output
—	61	GND	Ground
—	62	DL7n	Receiver inverted data output
—	63	DL7p	Receiver non-inverted data output
—	64	GND	Ground
—	70	GND	Ground
—	71	SL6p	Transmitter non-inverted data input
—	72	SL6n	Transmitter inverted data input
—	73	GND	Ground
—	74	SL4p	Transmitter non-inverted data input
—	75	SL4n	Transmitter inverted data input

Table 179C–4—MDI connector contact mapping for OSFP1600

OSFP1600	Connector signal name	Description
1	GND	Ground
2	SL1p	Transmitter non-inverted data input
3	SL1n	Transmitter inverted data input
4	GND	Ground
5	SL3p	Transmitter non-inverted data input
6	SL3n	Transmitter inverted data input
7	GND	Ground
8	SL5p	Transmitter non-inverted data input
9	SL5n	Transmitter inverted data input
10	GND	Ground
11	SL7p	Transmitter non-inverted data input
12	SL7n	Transmitter inverted data input
13	GND	Ground
18	GND	Ground
19	DL6n	Receiver inverted data output
20	DL6p	Receiver non-inverted data output
21	GND	Ground
22	DL4n	Receiver inverted data output
23	DL4p	Receiver non-inverted data output
24	GND	Ground
25	DL2n	Receiver inverted data output
26	DL2p	Receiver non-inverted data output
27	GND	Ground
28	DL0n	Receiver inverted data output
29	DL0p	Receiver non-inverted data output
30	GND	Ground
31	GND	Ground
32	DL1p	Receiver non-inverted data output
33	DL1n	Receiver inverted data output
34	GND	Ground
35	DL3p	Receiver non-inverted data output

Table 179C–4—MDI connector contact mapping for OSFP1600 (*continued*)

OSFP1600	Connector signal name	Description
36	DL3n	Receiver inverted data output
37	GND	Ground
38	DL5p	Receiver non-inverted data output
39	DL5n	Receiver inverted data output
40	GND	Ground
41	DL7p	Receiver non-inverted data output
42	DL7n	Receiver inverted data output
43	GND	Ground
48	GND	Ground
49	SL6n	Transmitter inverted data input
50	SL6p	Transmitter non-inverted data input
51	GND	Ground
52	SL4n	Transmitter inverted data input
53	SL4p	Transmitter non-inverted data input
54	GND	Ground
55	SL2n	Transmitter inverted data input
56	SL2p	Transmitter non-inverted data input
57	GND	Ground
58	SL0n	Transmitter inverted data input
59	SL0p	Transmitter non-inverted data input
60	GND	Ground

179C.2 MDI connector types

179C.2.1 QSFP224

For 200GBASE-CR1, 400GBASE-CR2, or 800GBASE-CR4, the mechanical interface between the PMD and the cable assembly may be a mated pair of QSFP224 plug and receptacle as illustrated in Figure 179C–1 and Figure 179C–2. If QSFP224 is used, the implementation shall comply with the QSFP224 component mechanical specification as defined in SFF-TA-1027, Rev 2.0, 2025-09-09, QSFP2 Connector, Cage, & Module Specification. QSFP224 supports up to four lanes.

The QSFP224 cable assembly connector is the plug illustrated in Figure 179C–1. The QSFP224 PMD connector is the receptacle with the mechanical mating interface as illustrated in Figure 179C–2. The

connectors have data signal and signal ground contact assignments as specified in Table 179C–3. The pin definition for both the plug and receptacle remains unchanged from Annex 162C.

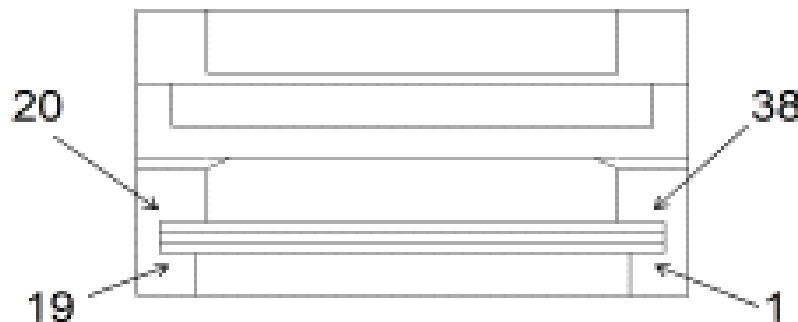


Figure 179C–1—QSFP224 cable assembly plug

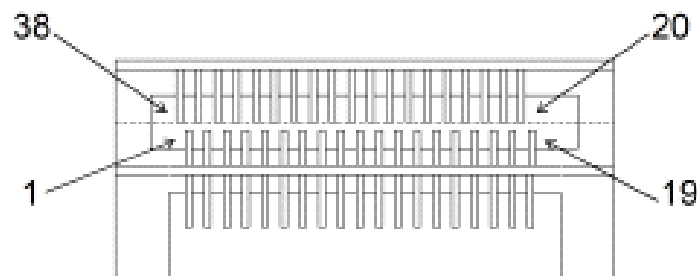


Figure 179C–2—QSFP224 PMD receptacle

179C.2.2 QSFP-DD1600

For 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, or 1.6TBASE-CR8, the mechanical interface between the PMD and the cable assembly may be a mated pair QSFP-DD1600 plug and receptacle as illustrated in Figure 179C–3 and Figure 179C–4. If QSFP-DD1600 is used, the implementation shall comply with the QSFP-DD1600 component mechanical specification as defined in the QSFP-DD/QSFP-DD800/QSFP-DD1600 Hardware Specification for QSFP Double Density 8x Pluggable Transceivers, Rev 7.1. QSFP-DD1600 supports up to eight lanes.

The QSFP-DD1600 cable assembly connector is the plug illustrated in Figure 179C–3. The QSFP-DD PMD connector is the receptacle with the mechanical mating interface as illustrated in Figure 179C–4. The connectors have data signal and signal ground contact assignments as specified in Table 179C–3.

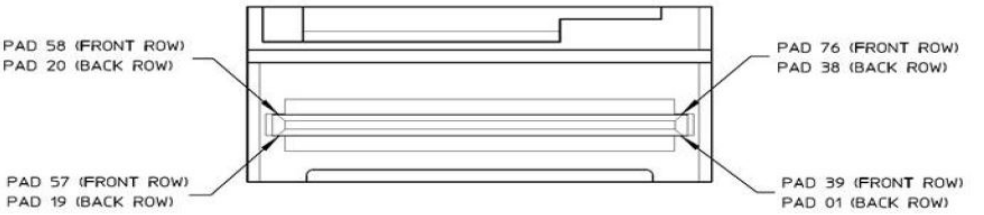


Figure 179C-3—QSFP-DD1600 cable assembly plug

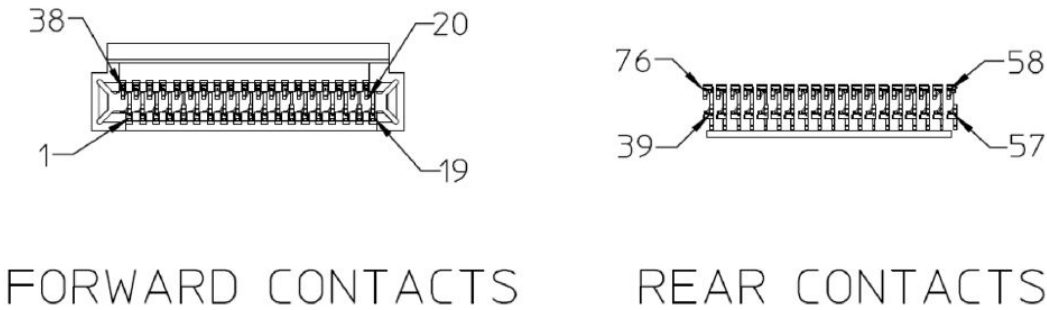


Figure 179C-4—QSFP-DD1600 PMD receptacle

179C.2.3 OSFP1600

For 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, or 1.6TBASE-CR8, the mechanical interface between the PMD and the cable assembly may be a mated pair of OSFP1600 plug and receptacle as illustrated in Figure 179C-5 and Figure 179C-6. If OSFP1600 is used, the implementation shall comply with the OSFP1600 component mechanical specification as defined in the OSFP Octal Small Form Factor Pluggable Module, Rev 5.22 specification. OSFP1600 supports up to eight lanes.

The OSFP1600 cable assembly connector is the plug illustrated in Figure 179C-5. The OSFP1600 PMD connector is the receptacle with the mechanical mating interface as illustrated in Figure 179C-6. The connectors have data signal and signal ground contact assignments as specified in Table 179C-4.

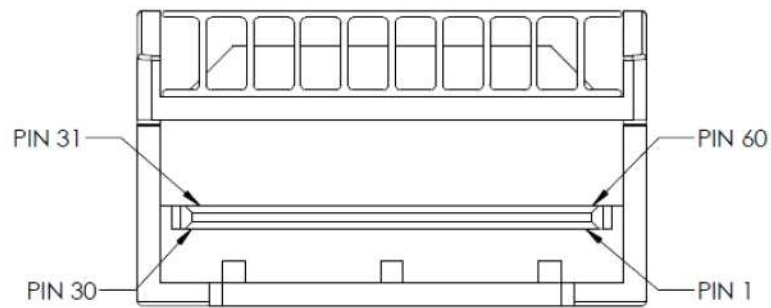


Figure 179C-5—OSFP1600 cable assembly plug

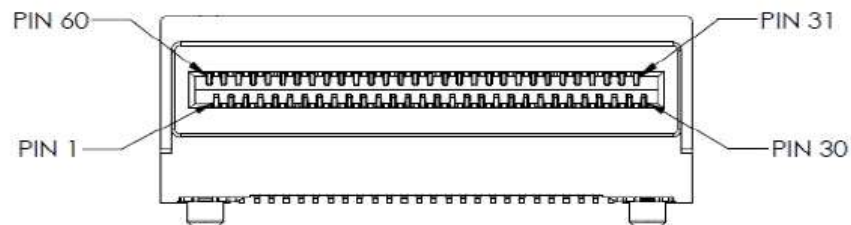


Figure 179C-6—OSFP1600 PMD receptacle

179C.3 Protocol implementation conformance statement (PICS) proforma for Annex 179C, MDIs for 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8¹⁰

179C.3.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Annex 179C, MDIs for 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8 shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

179C.3.2 Identification

179C.3.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1, 3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

179C.3.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-202x, Annex 179C, MDIs for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
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¹⁰Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

179C.3.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
*MDI1	QSFP224 MDI connector	179C.2.1		O	Yes [] No []
*MDI2	QSFP-DD1600 MDI connector	179C.2.2		O	Yes [] No []
*MDI3	OSFP1600 MDI connector	179C.2.3		O	Yes [] No []

179C.3.4 PICS proforma tables for MDIs for 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8

179C.3.4.1 Contact Mapping

Item	Feature	Subclause	Value/Comment	Status	Support
CM1	QSFP224 MDI connector contact mapping according to Table 179C–3	179C.2.1		MDI1:M	Yes [] N/A []
CM2	QSFP-DD1600 MDI connector contact mapping according to Table 179C–3	179C.2.2		MDI2:M	Yes [] N/A []
CM3	OSFP1600 MDI connector contact mapping according to Table 179C–4	179C.2.3		MDI3:M	Yes [] N/A []